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Future nutrient pollution poses a threat to Africa's freshwater fishes

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Abstract

Excessive levels of nutrients, particularly nitrogen and phosphorus, trigger eutrophication leading to harmful algal blooms and oxygen depletion that endangers freshwater fish species. Despite widespread awareness of these risks, efforts to protect freshwater fish species remain largely ineffective, especially under climate change and other anthropogenic pressures. Here, we assess present and future nutrient pollution risks across African river systems using two water quality models, Integrated Model to Assess the Global Environment-Global Nutrient Model (IMAGE-GNM) and Soil and Water Assessment Tool (SWAT+), under the SSP5–RCP8.5 scenario for 2010 and 2050. We evaluate total nitrogen and total phosphorus concentrations against ecological thresholds derived from fish metrics for freshwater fish species (0.7 mg N l^{-1} and 0.07 mg P l^{-1}). Both models project increasing nutrient pollution risks between 2010 and 2050 across major river basins and ecoregions, although SWAT+ consistently estimates higher impacts than IMAGE-GNM. By 2050, IMAGE-GNM projects that 42% of freshwater fish species will have at least half of their geographical range threatened by nutrient pollution, while SWAT+ projects similar high threats for more than 70% of species. Across all ecoregions, the potentially affected fraction of species is projected to increase between 2010 and 2050 with major river basins such as Niger, Volta and Turkana having over 80% of their fish species potentially affected by 2050. High risk conditions are particularly evident in habitat types such as temperate upland rivers, coastal rivers, and large lake basins. Our findings highlight regions where proactive management and policy interventions should be prioritized to mitigate the adverse effects of nutrient pollution on freshwater fish biodiversity in Africa.

1. Introduction

Nutrient pollution, driven predominantly by nitrogen (N) and phosphorus (P), is a critical environmental issue impacting water quality worldwide. Elevated levels of these nutrients lead to eutrophication, characterized by excessive algal blooms, diminished oxygen levels, significantly affecting aquatic ecosystems (Devlin and Brodie 2023). Sources of nutrient pollution include agricultural runoff, urban wastewater, and industrial discharges, all of which contribute to the degradation of freshwater and marine systems, with their impacts likely exacerbated by climate change (Suresh *et al* 2023). Nutrient consumption is increasing rapidly in developing countries, primarily due to population growth, with these economies projected to account for 75% of global nutrient use by 2050 (Sutton *et al* 2013). This poses significant

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challenges for Africa, where rapid population growth and economic development are outpacing the implementation of effective water management practices (Krishnadas and Matysik 2024). Studies have shown that upcoming shifts in water quality driven by socio-economic changes, climate change, and rising hydroclimatic extremes will significantly affect both people and ecosystems (Beusen *et al* 2022, van Vliet *et al* 2023), with Sub-Saharan Africa increasingly becoming the dominant hotspot of surface water pollution (Jones *et al* 2023).

The fishing industry is vital in Africa, supporting over 10 million people, including 7 million in West Africa alone (Westlund *et al* 2008). This sector significantly contributes to food security and economic livelihoods (Temesgen *et al* 2019). However, many regions are already experiencing adverse effects from nutrient enrichment such as impaired water quality and reduced biodiversity due to eutrophication and toxicity (Nyenje *et al* 2010, Zhou *et al* 2023). Pollution is recognized as one of the most significant threats to freshwater fish (Sayer *et al* 2025), often resulting in eutrophication which can lead to hypoxia. This process alters the water's physical and chemical properties by lowering pH levels to toxic thresholds for fish that impair their ability to grow, reproduce and survive (Mustafa *et al* 2024). Higher nutrient levels promote algal blooms which affect both uptake and supply of dissolved oxygen (Camargo and Alonso 2006, Yan *et al* 2021). These blooms reduce aquatic life's ability to find food by blocking sunlight, potentially driving species away or causing local extinctions (Villacorte *et al* 2015). Additionally, they can clog fish gills and produce toxins that lead to illness or death in fish (EPA 2015). As a key indicator of ecosystem health (Villéger *et al* 2017), numerous studies have evaluated the impacts of nutrient pollution on fish biodiversity in various regions worldwide (Tilak *et al* 2007, Gaber *et al* 2013, Wagenaar and Barnhoorn 2018, Zhou *et al* 2024). However, the effects of N and P pollution on freshwater fish biodiversity in tropical regions are rarely studied (Dudgeon and Strayer 2024), with studies lacking in spatial and temporal coverage, especially in Africa.

Climate change and rapid socio-economic development undermine efforts to halt biodiversity loss in freshwater ecosystems already under significant stress (Birk *et al* 2020, Capon *et al* 2021). More specifically, climate change is currently threatening 50% of the world's freshwater fish species (Ahmed *et al* 2022). Moreover, it intensifies other already established threats, such as water quality degradation and river fragmentation (Dudgeon and Strayer 2024). It also alters the flows in streams and rivers and even if these do not dry out, reduced flows can lead to increased pollution concentrations or give rise to toxic algal blooms (Whitehead *et al* 2009). On the other hand, high flows mobilize nutrients, sediments, toxic chemicals and aid in dispersal of non-native species (Dudgeon and Strayer 2024). Abrupt changes in flow or temperature in rivers can substantially reduce fish species richness (Sabater *et al* 2023), with decreased water levels inhibiting fish growth (Sanon *et al* 2020). Therefore, as climate change impacts on freshwater biodiversity are projected to intensify (Ahmed *et al* 2022), it is important to assess the potential threat of future nutrient pollution on freshwater fish.

To effectively manage and mitigate nutrient pollution, it is crucial to establish future hotspots where nutrient levels are likely to rise. This foresight can enable detailed studies in the hotspot regions for targeted interventions and policy planning. In this sense, scenario-based projections offer valuable insights into how nutrient pollution might evolve under different socio-economic and climatic conditions (O'Neill *et al* 2017). In this study, we focus on the high-end warming scenario SSP5-RCP8.5, characterized by high greenhouse gas emissions and rapid economic growth (Riahi *et al* 2011, Kriegler *et al*, 2017). We employed two large scale water quality models namely the Integrated Model to Assess the Global Environment—Global Nutrient Model (IMAGE-GNM) (Beusen *et al* 2022), and Soil and Water Assessment Tool (SWAT+) (Nkwasa *et al* 2024) to project future nutrient pollution hotspots across Africa. Models are increasingly used to address data gaps in water research, especially in regions lacking water quality data and robust monitoring programs, such as Sub-Saharan Africa (Desbureaux *et al* 2022). Importantly, these models are capable of simulating the impacts of climate change on future water quality (Strokal *et al* 2024). Since changes in water quality will not only affect human water use (Jones *et al* 2022), but also freshwater ecosystems (Bosmans *et al* 2022), it is essential to investigate the risk they pose on freshwater fish species.

In this study, we assessed the threat of future nutrient pollution to freshwater fish species by first comparing total nitrogen (TN) and total phosphorus (TP) concentrations from two models: IMAGE-GNM and SWAT+. This comparison, focused on identifying nutrient pollution hotspots, based on simulations under the combined climate change and socioeconomic scenario SSP5-RCP8.5 for the years 2010 and 2050. Utilizing a regional dataset of 2624 freshwater fish species (Snoeks and Theeten 2025), we estimated the threat posed by elevated TN and TP levels to fish in different ecoregions and major river basins. This study is, to our knowledge, the first assessment of potential threats of future nutrient pollution on riverine African freshwater fish. These insights into hotspots for nutrient pollution are

crucial for policymakers and stakeholders to safeguard water quality, protect aquatic ecosystems and to decide on policy interventions.

2. Materials and methods

2.1. Water quality models

2.1.1. IMAGE–GNM

IMAGE-GNM is a spatially explicit, dynamic model with a resolution of 0.5° × 0.5° that has been validated with sampling station data. It simulates the pathways of N and P to surface water from concentrated point sources, such as wastewater from industrial, urban, and rural populations, as well as from diffuse sources like agricultural systems with fertilizer use and manure management at a yearly time step. Additionally, it considers natural ecosystems, surface runoff, shallow and deep groundwater, riparian zones, litterfall in floodplains, and atmospheric deposition (Beusen *et al* 2015b). IMAGE–GNM uses the grid-based global hydrological model PCRaster global water balance (PCR-GLOBWB) (Sutanudjaja *et al* 2018), to quantify water stores and fluxes, volume, surface area, and thus depth of water bodies, and water travel time. In the model, grid cells receive water containing N and P from upstream cells, and N and P entering water bodies through diffuse, point and natural sources. Nutrient retention is calculated based on water residence time and nutrient uptake velocity before water and nutrients are transported downstream. In this study, we used model estimates of river TN and TP for 2010, the latest non-scenario year, and for 2050 under the SSP5-RCP8.5 scenario. The model was validated against observed discharge, N and P concentrations in a number of different stations in a variety of rivers (Beusen *et al* 2022). The data used in this study was taken from its updated version, the IMAGE 3.2 version detailed in Beusen *et al* (2022), incorporating enhancements such as new input data, and an improved PCR-GLOBWB model with updated hydrological data.

2.1.2. SWAT+ model

SWAT+ is an enhanced version of the SWAT model, developed to offer greater flexibility in connecting spatial units and modeling management practices (Bieger *et al* 2017, Arnold *et al* 2018). The semi-distributed model simulates hydrological dynamics, crop growth, sediment transport, and nutrient loads by dividing hydrological basins into sub-basins and hydrologic response units (HRUs), with each HRU representing areas of similar land use, soil type, slope, and management practices (Neitsch *et al* 2005). SWAT+ also distinguishes upland and lowland processes through landscape units and uses the water balance concept as the foundation for all hydrological processes. The model simulates N and P cycles, tracking both inorganic and organic forms of these nutrients. Nitrogen movement and transformation are calculated based on water volume and soil nitrate concentration, while organic N and P transport via sediment is estimated using topsoil concentrations, sediment yield, and an enrichment ratio. The enrichment ratio accounts for the difference between the nutrient content of eroded sediment and the original soil, as finer particles transported during erosion tend to have higher nutrient concentrations compared to bulk soil. Nitrogen enters the soil through fertilizer and manure application, atmospheric deposition, and nitrogen fixation, while removal occurs via plant uptake, volatilization, denitrification, percolation, tile flow, lateral flow, and runoff. Phosphorus mainly enters through fertilizer and manure application and is removed by plant uptake, tile flow, lateral flow, percolation, and runoff.

Nutrients from the land phase can enter channels and move downstream through surface runoff and lateral subsurface flow (Neitsch *et al* 2011). In-stream nutrient dynamics are simulated using the kinetic routines from the QUAL2E in-stream water quality model (Brown 1987). The model tracks both nutrients dissolved in the stream and those attached to sediment particles. Dissolved nutrients travel with the water flow, while sediment-bound nutrients can settle onto the channel bed with the sediment. Additional information on nutrient processes in SWAT+ can be found in the SWAT documentation (Arnold *et al* 1998, Neitsch *et al* 2011). Details on the setup, calibration and validation of the Africa SWAT+ model using observed river discharge data, TN and TP concentrations, and river sediment loads, are provided in Nkwasa *et al* (2024).

2.2. Future socio-economic and climate scenarios

Shared socioeconomic pathways (SSPs) and representative concentration pathways (RCPs) describe future possible developments in socio-economic and climatic conditions, respectively, offering integrated scenarios to explore uncertain future changes. Consistent scenario assumptions are important for enabling meaningful comparison of model outputs across different scales and variables. In this study, all models were driven using harmonized socio-economic assumptions based on SSP5 together with climate forcing

from RCP8.5 to simulate future water quality conditions. SSP5 is characterized by high socio-economic growth with heavy fossil fuel use and increased high-tech agricultural production (O'Neill *et al* 2017), whereas RCP8.5 considers a high level of greenhouse gas emissions (van Vuuren *et al* 2011). The scenario assumes intensive agricultural practices to meet the growing food demand, with a diet rich in meat leading to increased manure production. This pathway projects significant economic and population growth, with Africa's gross domestic product expected to rise by 60% and its population increasing by 70% to reach 1.7 billion people by 2050 (Samir and Lutz 2014). Despite the increase in sewage connections, inadequate wastewater treatment with poor environmental management policies will increase the risks to water quality and aquatic biodiversity. In addition, under SSP5–RCP8.5, the energy-intensive lifestyle and associated greenhouse gas emissions lead to strong climate changes (Siabi *et al* 2023), which affect hydrological conditions and nutrient flows. Generally, this scenario envisions a future marked by urbanization, low-ambition nitrogen policies, improved sewage connections with limited advancements in treatment, high water withdrawal, and inadequate climate mitigation efforts (Wang *et al* 2024). This high emissions scenario provides insights into the potential scale of impacts on ecosystems under extreme pressure from both climate change and socio-economic drivers. This makes it particularly relevant for identifying nutrient pollution hotspots and assessing resulting threats to freshwater fish species, thereby highlighting the urgency for management and policy actions in the region.

IMAGE-GNM nutrient simulations were performed at a spatial resolution of 0.5° whereas the SWAT+ nutrient simulations were initially performed at HRU level and then aggregated from the sub-basin scale to a 0.5° for comparative analysis Click or tap here to enter text. An overview of the major assumptions and implementations for both IMAGE-GNM and SWAT+ models is provided in tables S1 and S2, supplementary material.

2.3. Thresholds for TN and TP

Numerous nutrient concentration thresholds for TN and TP have been established across various regions and water bodies to protect biodiversity, maintain good ecological conditions and mitigate eutrophication. These efforts aim to preserve water quality and prevent ecological damage caused by nutrient pollution. Table S3 (supplementary material) summarizes nutrient thresholds linked to fish fauna, benthic macroinvertebrates, harmful algal blooms, excessive cyanobacterial growth, and unwanted periphyton blooms in aquatic systems. For our study, we selected thresholds based on the ecological effects of nutrients on river systems, particularly concentrations beyond which fish assemblages are likely to experience significant degradation. Therefore, we adopted thresholds derived from fish-related metrics. In the absence of Africa-specific thresholds, we relied on general freshwater thresholds as a reasonable starting point for the risk assessment.

Several studies have identified nutrient thresholds based on their effects on fish communities. For example, Azevedo *et al* (2013) proposed a TP threshold of 0.07 mg P l^{-1} for freshwater fish species richness in sub(tropical) streams globally while Wang *et al* (2007) identified mean thresholds where fish metrics change most dramatically at 0.07 mg l^{-1} TP and 0.9 mg l^{-1} TN. Robertson (2006) reported breakpoints in fish indices at 0.06 mg l^{-1} TP and 0.54 mg l^{-1} TN in Wisconsin rivers whereas Miltner and Rankin (1998) proposed a similar threshold of 0.06 mg l^{-1} TP and 0.61 mg l^{-1} total inorganic nitrogen, beyond which fish community structure is significantly degraded based on the Index of Biotic Integrity. These fall within relatively narrow ranges of 0.54–0.90 mg l^{-1} TN and 0.06–0.07 mg l^{-1} for TP reported for fish fauna. Based on the above previous studies, we adopted TN and TP thresholds of 0.7 mg N l^{-1} and 0.07 mg P l^{-1}, respectively for our study. These values align with trophic transitions for streams, such as the mesotrophic–eutrophic boundary for TP (0.075 mg l^{-1}) and the oligotrophic–mesotrophic boundary for TN (0.700 mg l^{-1}) (Dodds *et al* 1998) and are also broadly consistent with values protecting ecosystems from eutrophication where most studies indicate TP threshold of 0.03–0.06 mg P l^{-1} and a TN threshold of 0.5–2.0 mg N l^{-1}, beyond which algal biomass can reach nuisance levels, and diatom composition undergoes significant changes (Poikane *et al* 2021). These values are also comparable to those derived in previous studies using other ecological indicators (see table S3, supplementary material) and provide meaningful endpoints for protecting aquatic life in large rivers.

2.4. Model intercomparison

We compared annual IMAGE-GNM and SWAT+ TN and TP simulations using two approaches: (1) analysing the projected percentage changes in TN and TP concentrations across Africa from 2010 to 2050 and (2) evaluating the percentage agreement between the models in classifying grid cells as critical or non-critical based on the ecological thresholds adopted in this study (0.7 mg N l^{-1} for TN and 0.07 mg P l^{-1} for TP).

2.5. Assessing threat to freshwater fishes

Freshwater fish occurrence data was sourced from the Global Biodiversity Information Facility (GBIF) which hosts the Royal Museum for Central Africa's collection, one of the largest repositories of African freshwater fish in the world (Snoeks and Theeten 2025). Initially, we retrieved 2719 freshwater fish species with 86 415 occurrence records. After data cleaning, which involved removing records lacking species names or geographic information and eliminating duplicates, our final dataset comprised 2624 freshwater fish species with 74 025 occurrence records dating from 1896 to 2023. Lentic species were excluded from the study, as the models used are optimized solely for river systems. The complete dataset is available in the supporting information.

Organisms generally adapt to climate change by migrating to more suitable habitats (Loarie *et al* 2009). However, predicting such movements for freshwater fish is challenging due to uncertainties and data gaps related to current and future barriers, such as dams and weirs (McManamay *et al* 2018). Additionally, reliable data on dispersal abilities are unavailable for most species (Radinger *et al* 2017). To address these limitations, we adopted the 'no dispersal' assumption, restricting fish species to their current geographical ranges within the grid cells where they are presently located (Barbarossa *et al* 2021). Due to limited data on fish population decline rates, our assessments relied on presence/absence of data over time, assuming uniform abundance within species ranges and linear rates of decline. While we acknowledge the importance of lower thresholds to prevent nutrient deficiency in fish species, this study focused on high nutrient pollution extremes. We assume that fish species can tolerate nutrient-induced hypoxia or toxicity up to the specified TN and TP thresholds of 0.7 mg N l^{-1} for TN and 0.07 mg P l^{-1} for TP explained above, with uniform tolerance across species within each ecoregion or habitat.

To evaluate the potential threat of nutrient pollution to freshwater fish species, we first identified and mapped grid cells where annual TN and TP concentrations from the two water quality models exceeded the critical thresholds. We then created a combined exceedance map for each model indicating areas where at least one of the two nutrients surpassed its critical threshold. Following methodology from Barbarossa *et al* (2021), for each species, we quantified the proportion of its geographical range exposed to nutrient pollution by flagging grid cells as 'threatened' if either TN and/or TP exceeded their respective thresholds. The percentage of geographical range threatened (PGT) for each species (x) was calculated as:

$$\mathrm{PGT}_x = \frac{\mathrm{AT}_x}{\mathrm{A}_x} \times 100$$

where AT_x is the total number of grid cells where species x is present and at least TN and/or TP threshold is exceeded, and A_x is the total number of grid cells where species x is present.

To assess the proportion of species threatened by nutrient pollution in each ecoregion, we computed the potentially affected fraction (PAF) of freshwater species in 2010 and 2050. A freshwater ecoregion is a large area encompassing one or more freshwater systems that contains a distinct assemblage of natural freshwater communities and species, where the freshwater species, dynamics, and environmental conditions within the ecoregion are more similar to each other than to those of surrounding ecoregions, forming a conservation unit (FEOW 2025). The PAF for each ecoregion was determined as follows,

$$\mathrm{PAF}_{\mathrm{ecoregion}} = \frac{\mathrm{ST}_{\mathrm{ecoregion}}}{\mathrm{S}_{\mathrm{ecoregion}}}$$

where $\mathrm{PAF}_{\mathrm{ecoregion}}$ represents the PAF of freshwater fish species in an ecoregion, $\mathrm{ST}_{\mathrm{ecoregion}}$, represents the number of unique species in grid cells within the ecoregion where at least one critical threshold (TN and/or TP) is exceeded, and $\mathrm{S}_{\mathrm{ecoregion}}$ represents the total number of unique species in the ecoregion. The computation carefully avoids repetition if the same species appear in more than one threatened grid cell. A similar analysis was also done for the major river basins in Africa.

3. Results

3.1. Model intercomparison for nutrient hotspots

Projected changes in nutrient concentrations showed strong spatial variability across Africa, with both IMAGE-GNM and SWAT+ projecting widespread increases in TN and TP concentrations by 2050 under SSP5–RCP8.5 (figure 1). IMAGE-GNM projected that over 80% of Africa's rivers will experience higher TN and TP concentrations in 2050 relative to 2010, although localized decreases were observed in parts of southern and central Africa. Similarly, SWAT+ projected increases in TN concentrations across

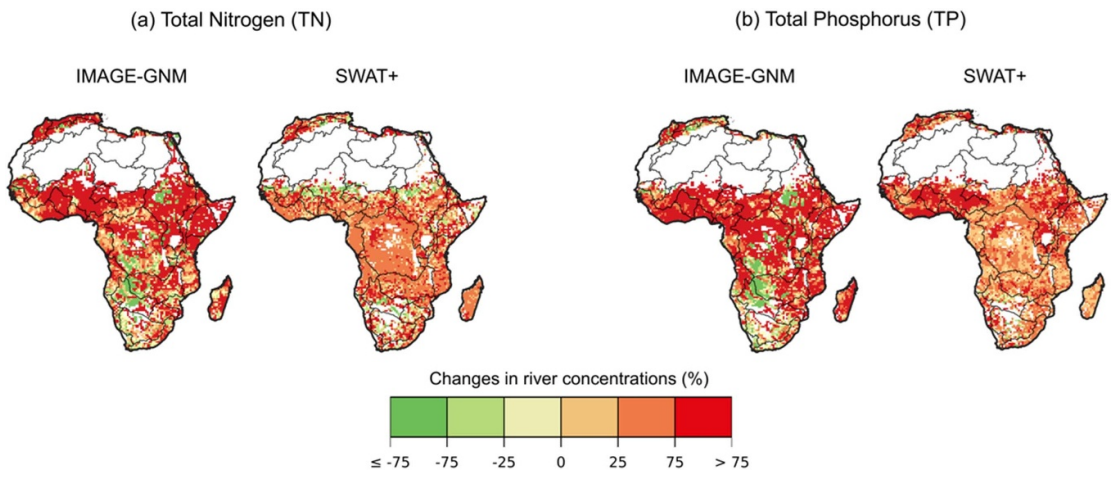


Figure 1. Relative percentage change in (a) total nitrogen (TN) and (b) total phosphorus (TP) river concentrations between 2010 and 2050 based on the SSP5-RCP8.5 scenario for the IMAGE-GNM and SWAT+ model. Polygons represent river basins and white areas represent regions with no available model data (e.g. non-river cells or cells excluded during masking).

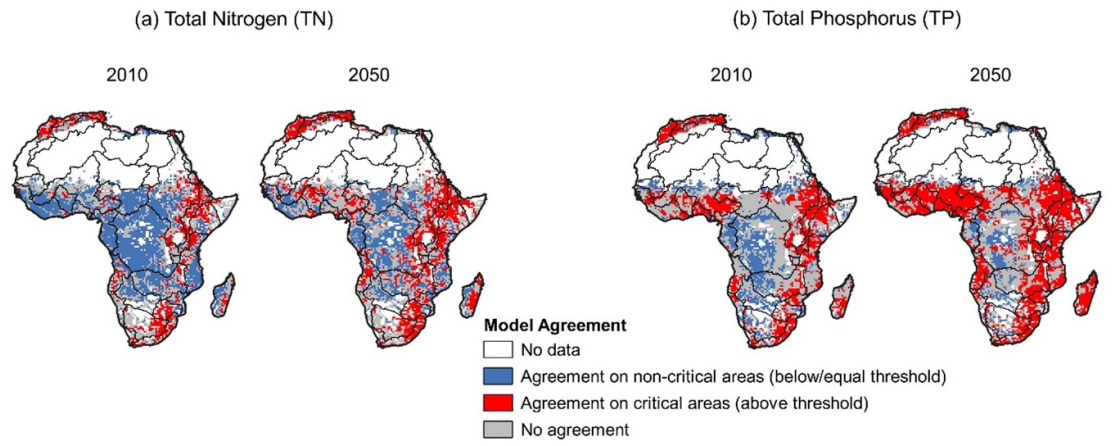


Figure 2. Spatial agreement between IMAGE-GNM and SWAT+ models in classifying nutrient pollution risk across Africa for (a) total nitrogen (TN) and (b) total phosphorus (TP) in 2010 and 2050 under SSP5-RCP8.5 according to predefined thresholds for freshwater fish species: 0.7 mg N l^{-1} for TN and 0.07 mg P l^{-1} for TP. Polygons represent river basins and white areas represent regions with no available model data (e.g. non-river cells or cells excluded during masking).

approximately 80% of rivers and increases in TP concentrations across about 95% of rivers. Compared to IMAGE-GNM, SWAT+ showed less extensive areas with decreasing nutrient concentrations. Similar patterns are observed at the river basin scale (figure S1), where several basins in eastern, western, and southern Africa consistently showed increasing TN and TP concentrations.

Spatial agreement between IMAGE-GNM and SWAT+ in classifying critical and non-critical nutrient pollution areas shows clear spatial and temporal variability across Africa (figure 2). For TN, overall model agreement decreases from 66% in 2010 to 52% in 2050. Areas consistently classified as critical, exceeding nutrient thresholds by both models, particularly in eastern and southern Africa, increased from 10% to 27%, while robust non-critical areas declined from 56% to 25%. Model disagreement also increased from 34% to 48%, indicating greater uncertainty in the future spatial distribution of nitrogen hotspots. For TP, overall agreement increased slightly from 64% in 2010 to 68% in 2050. Agreement on critical hotspot areas increased from 18% to 32%, suggesting an expansion of regions consistently identified as phosphorus hotspots by both models. In contrast, robust non-critical areas declined from 46% to 36%, while model disagreement decreased slightly from 36% to 32%. Overall, both models project an expansion of nutrient pollution hotspots by 2050, although future TP projections show slightly stronger consistency between models than TN projections. Similar patterns are observed at the river basin scale (figure S1), where both models consistently classify several major basins in eastern and southern Africa as critical, particularly for TP. In contrast, several basins in western and central Africa remained non-critical or showed disagreement between the models, especially for TN.

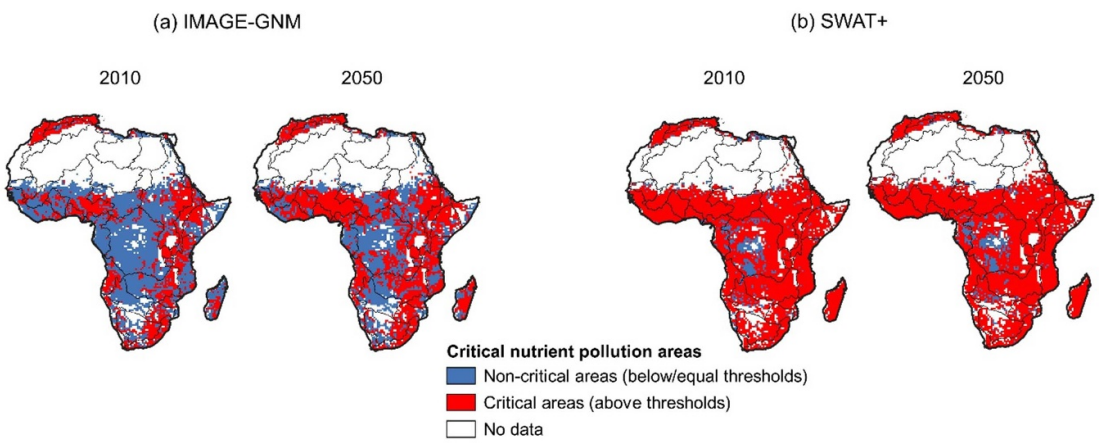


Figure 3. Spatial distribution of critical nutrient pollution areas across Africa for (a) IMAGE-GNM and (b) SWAT+ in 2010 and 2050 under SSP5–RCP8.5, based on combined exceedance of TN and/or TP thresholds for freshwater fish species (0.7 mg N l^{-1} for TN and 0.07 mg P l^{-1} for TP). Red areas indicate grid cells where at least one nutrient threshold was exceeded, while blue areas represent non-critical areas where both nutrient concentrations remained below or equal to the thresholds. Polygons represent river basins and white areas indicate regions with no available model data or non-river cells excluded during masking.

3.2. Threat of species exposure to nutrient pollution

IMAGE-GNM projected that the critical areas with combined TN and/or TP exceedance would increase from 38% in 2010 to 59% in 2050, while non-critical areas reduce from 62% to 41% (figure 3(a)). In contrast, SWAT+ projected more spatially extensive affected areas, with critical areas already covering about 90% of Africa's river systems in 2010 and increasing slightly to 91% by 2050. Correspondingly, non-critical areas in SWAT+ remained below 10% for both periods. Spatially, both models identified extensive critical areas in eastern, western, and southern Africa, with hotspot regions becoming more widespread by 2050. However, IMAGE-GNM showed larger proportions of non-critical regions, particularly across central Africa, whereas SWAT+ projected more continuous and spatially extensive exceedance patterns. Overall, these differences suggest that SWAT+ projects broader nutrient threshold exceedance across African rivers, while IMAGE-GNM indicates stronger spatial heterogeneity and localized non-critical regions. Similar patterns were also observed at the river basin scale (figure S3).

Our results indicate a substantial increase in the proportion of freshwater fish species with their geographical ranges threatened with future nutrient pollution (figure 4(A)). Based on IMAGE-GNM projections, the percentage of species with at least half of their geographical range threatened increased from 15% in 2010 to 42% by 2050. On the other hand, species with no threatened range decreased from 60% to 35%, indicating a marked expansion of nutrient pollution prone areas across African freshwater ecosystems. In contrast, SWAT+ projected consistently higher risks of nutrient pollution, with over 70% of fish species having at least half of their geographical range threatened in 2010 and 2050.

Several fish orders, including Synbranchiformes (swamp eels), Clupeiformes (freshwater herrings and sardines e.g. *Limnothrissa, Stolothrissa*), Cypriniformes (*Labeo, Enteromius*), and Siluriformes (catfishes), showed high vulnerability to nutrient pollution, particularly under the SWAT+ model, where more than 80% of their geographical range was threatened in both 2010 (figure 4(B)) and 2050 (figure 4(C)). Similarly, Polypteriformes (bichirs) and Characiformes (African tetras and tigerfishes) exhibited substantial increases in threatened range by 2050. In contrast, IMAGE-GNM projected comparatively lower threatened ranges across most fish orders, especially in 2010. Orders such as Gonorynchiformes (kneriids, hingemouths, and sandfishes), Anguilliformes (freshwater eels), and Mugiliformes (mullets) showed relatively lower threatened ranges, generally remaining below 50% under IMAGE-GNM projections. Overall, both models projected an increase in the geographical range threatened by nutrient pollution between 2010 and 2050, although SWAT+ consistently estimated substantially higher impacts than IMAGE-GNM across nearly all fish orders.

The ecoregions database indicates that central and western Africa, characterized by (sub)tropical floodplain rivers, coastal rivers and wetlands have the highest freshwater fish species richness. In contrast, southern and northern regions, dominated by xeric freshwaters, closed basins, and Montane freshwaters, exhibit the lowest species diversity due to limited water availability (figure S5,a–b). Both IMAGE-GNM and SWAT+ projected increasing PAF values for freshwater fish species from 2010 to 2050 across most ecoregions, indicating greater future eutrophication risks (figure 5). Particularly high values were observed in temperate upland rivers, temperate coastal rivers, and large lake basins, where PAF values

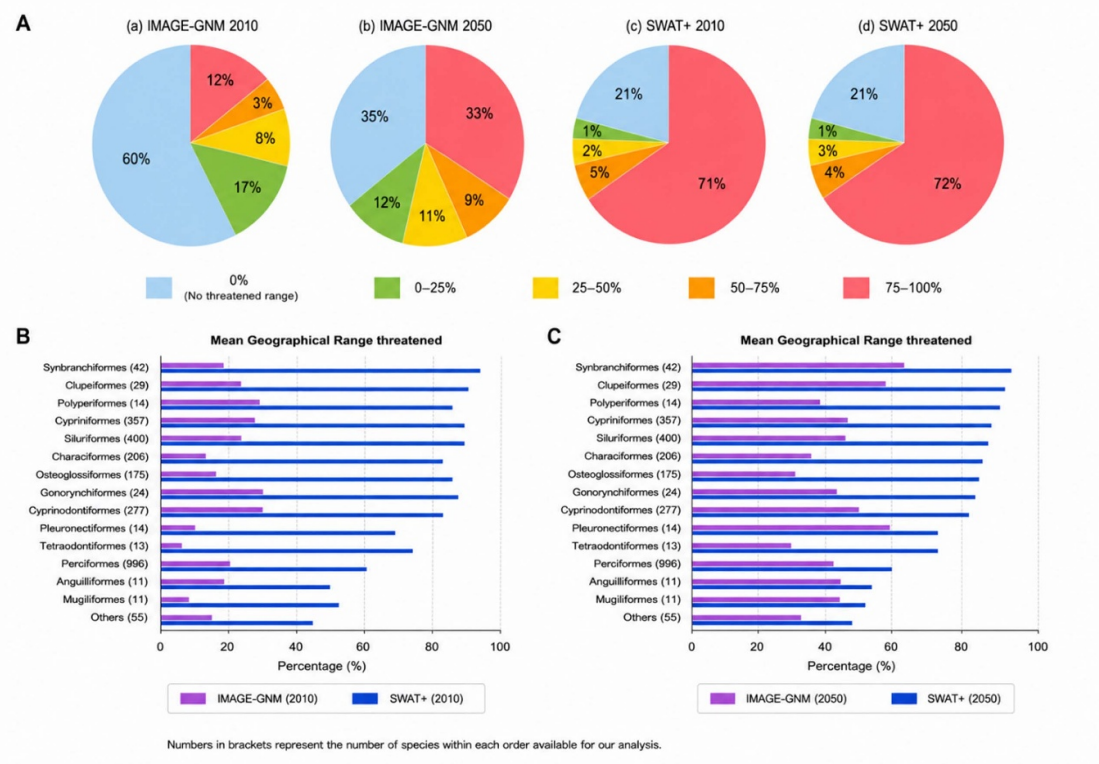


Figure 4. (A) Total percentage of freshwater fish species (numbers in the pie-chart) that fall within different percentage classes of geographical range threatened by nutrient pollution for IMAGE-GNM in (a) 2010 and (b) 2050 and for SWAT+ in (c) 2010 and (d) 2050. (B) Percentage of geographical range threatened by nutrient pollution for different riverine fish orders in Africa in 2010 and (C) 2050. Numbers in brackets represent the number of species within each order available for our analysis. Fish orders in the group 'Others' are those with less than 10 unique species. These can be found in table S4, supplementary material. Source data is provided as supplementary information.

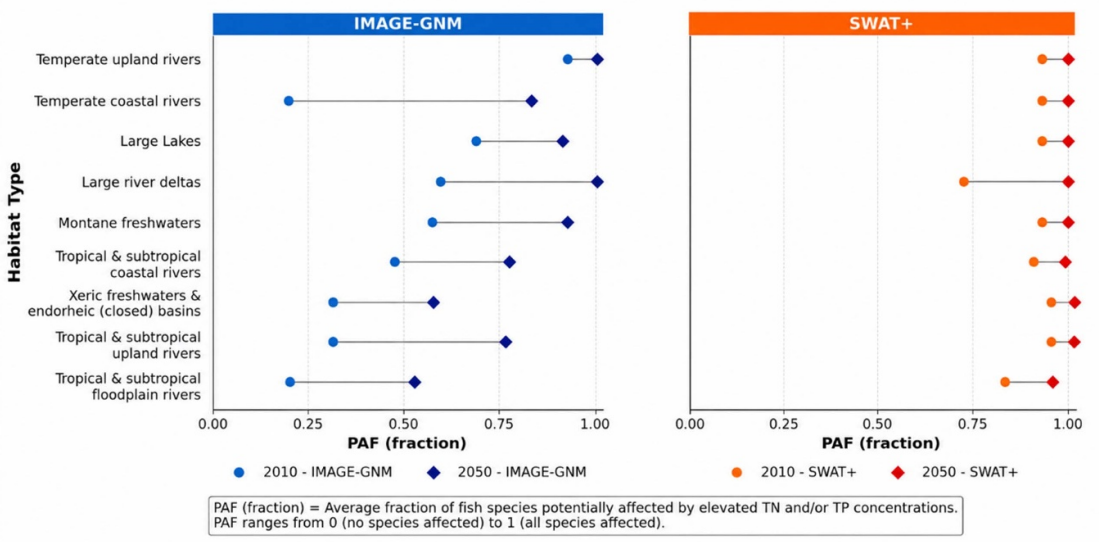


Figure 5. Average potentially affected fraction (PAF) of freshwater fish species across different freshwater habitat types in Africa for 2010 and 2050, comparing results from the IMAGE-GNM and SWAT+ models. Horizontal lines connect the two time periods for each habitat type, illustrating projected changes in biodiversity exposure to nutrient pollution.

approached or reached 1.0 by 2050. In contrast, tropical and subtropical floodplain rivers and xeric freshwaters and endorheic basins showed relatively lower values especially with IMAGE-GNM, although notable increases were still projected.

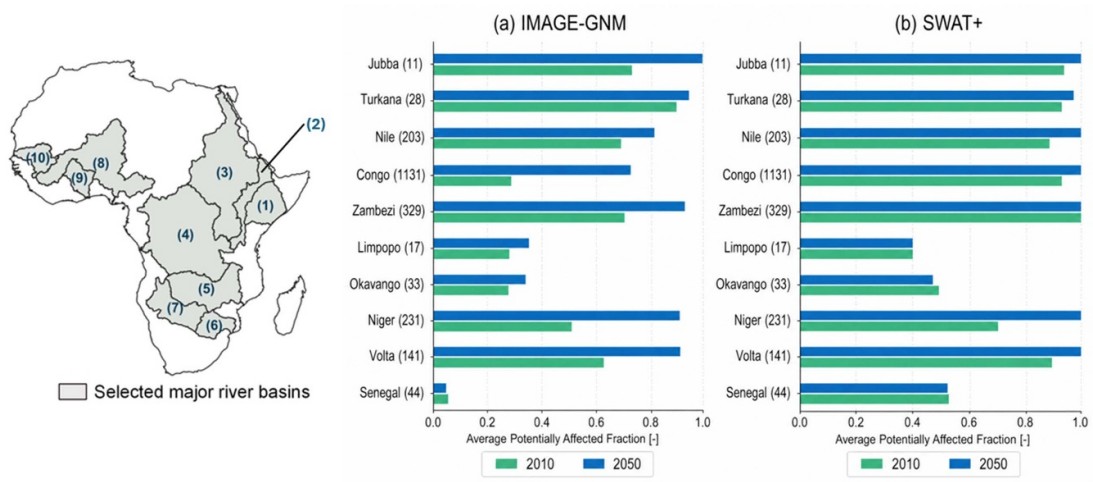


Figure 6. Left: spatial delineation of the selected major African river basins: (1) Jubba, (2) Turkana, (3) Nile, (4) Congo, (5) Zambezi, (6) Limpopo, (7) Okavango, (8) Niger, (9) Volta, and (10) Senegal. **Right:** average potentially affected fraction (PAF) of freshwater fish species in each selected river basin exposed to elevated TN and/or TP concentrations simulated by (a) IMAGE-GNM and (b) SWAT+ models exceeding the defined ecological thresholds for the years 2010 and 2050 under SSP5–RCP8.5. Numbers in brackets indicate the total number of freshwater fish species included in the analysis for each basin.

At the river basin scale, we calculate that PAF will increase in all river basins (figure S6). Among the selected basins, the Jubba, Turkana, Nile, and Zambezi basins exhibited particularly high PAF values in both models, with several projections approaching or reaching 1.0 by 2050, suggesting that nearly all freshwater fish species within these basins could potentially be exposed to nutrient concentrations exceeding ecological thresholds (figure 6). The Congo Basin, despite containing the highest number of freshwater fish species (1131 species), showed moderate PAF values in IMAGE-GNM but substantially higher values in SWAT+, especially in 2050. Relatively lower PAF values were observed in the Limpopo, Okavango, and Senegal basins, although increasing trends were still projected over time. In general, both IMAGE-GNM and SWAT+ projected increasing PAF values from 2010 to 2050 under SSP5–RCP8.5, indicating rising eutrophication risks across major African river systems. SWAT+ generally projected higher PAF values than IMAGE-GNM, particularly in the Niger and Volta basins, where future conditions approached near-complete freshwater fish species exposure to elevated nutrient concentrations.

4. Discussion

4.1. Model intercomparison on a continental scale

Our results show that both SWAT+ and IMAGE-GNM models consistently project increased TN and TP concentrations in African rivers by 2050 under SSP5-RCP8.5. The models show moderate agreement in identifying pollution hotspots which suggests that they can provide useful insights into eutrophication risks under this scenario. Despite broadly similar spatial patterns of nutrient increases, the magnitude of these changes differs between the models. There are several possible reasons for these differences. Table S1 gives an overview of the main assumptions and implementations of SWAT+ and IMAGE-GNM. IMAGE-GNM projects that ∼55% of rivers will have an increase in TN or TP greater than 75%, which is mainly explained by the dynamic land use change. The expansion of agricultural land and the reduction of natural land cover is causing relative high changes (in percentage). SWAT+ uses a constant land use map of 2010, thus the agricultural increase is more limited (in relative sense). In areas where grassland is playing an important role (for example Europe, but not Africa), the difference between the models could be more significant. Also, IMAGE-GNM explicitly accounts for long-term nutrient legacies from historical groundwater accumulation, whereas SWAT+ simulates groundwater flow and nitrate transport but does not explicitly represent nutrient legacies from historical loads. Additionally, IMAGE-GNM incorporates aquaculture as an anthropogenic nutrient source which is not represented in SWAT+.

Differences in hydrology between the two models influence not only the magnitude of river discharge (figure S10) and nutrient transport, but also nutrient timing and residence time, contributing to contrasting future projections. For example, IMAGE-GNM is sensitive to structural uncertainties in its underlying hydrological model, PCR-GLOBWB (Sutanudjaja *et al* 2018). In our study, SWAT+ is driven by an ensemble of four CMIP5 GCMs (MIROC5, IPSL-CM5A-LR, HadGEM2-ES, and GFDL-ESM2M),

with distinct precipitation and temperature biases (Sierra *et al* 2015, Liu *et al* 2022), whereas IMAGE-GNM uses the HadCM3 model, which tends to overestimate precipitation in some regions (Valdes *et al* 2017). Such differences in climate forcing can substantially influence hydrological responses, nutrient leaching, and runoff generation, ultimately leading to contrasting TN and TP projections. In addition, IMAGE-GNM explicitly represents lakes and reservoirs and accounts for nutrient retention during routing (Beusen and Bouwman 2022), whereas the current SWAT+ setup does not explicitly include reservoir and lake trapping processes (Teran *et al* 2026), which may lead to higher downstream nutrient export estimates given that reservoirs can retain approximately 20%–25% of continental nitrogen and phosphorus loads (Beusen *et al* 2015a). Nevertheless, despite these structural differences, the main assumptions that can influence water quality are almost the same for both models hence why both models identify comparable large-scale nutrient pollution hotspots across Africa.

These two models, to our knowledge, are the only existing spatially distributed nutrient models, making them the only suitable option for our study. Both models consistently identify nutrient pollution hotspots such as the Nile and Niger basins and parts of Northern and Southern Africa, offering a foundation for prioritizing regions for water quality monitoring (Pérez *et al* 2017). However, the model discrepancies in hotspot identification emphasize the need for model intercomparison frameworks to better evaluate these models (van Vliet *et al* 2019). Reliance on single model predictions can increase susceptibility to uncertainty, therefore using at least two models allows for model intercomparison and increases confidence in identifying robust patterns of nutrient pollution, particularly where both models consistently identify the same hotspot regions and show high spatial agreement in threshold exceedance. The ISIMIP water quality community, for example, established a protocol to harmonize model inputs enabling cross-scale assessments under climate change (Strokal *et al* 2024). From this, models can identify pollution hotspots, sources, and trends, particularly in data-scarce regions like Africa with limited monitoring networks.

4.2. Nutrient pollution and its threat to freshwater fish species

Our analysis reveals a significant projected increase in nutrient pollution across African rivers under the SSP5-RCP8.5 scenario by 2050, largely driven by population growth and increasing food demand (Wang *et al* 2024). The results showed that based on IMAGE-GNM projections, rivers exceeding at least one of the nutrient thresholds increase from 38% in 2010 to 59% in 2050, indicating intensified nutrient loading predominantly from anthropogenic sources (Beusen and Bouwman 2022). Agricultural intensification with the use of synthetic fertilizers and urbanization are key contributors to this trend (Beusen *et al* 2022). For instance, Soro *et al* (2023) demonstrated that nutrient enrichment in Côte d'Ivoire's major rivers stems predominantly from agricultural sources, particularly fertilizer use. This pattern is common across Africa, where agricultural lands often lie close to river waters (Thomas and Whittington 2023). On the other hand, Yasin *et al* (2010) reported a shift over time from natural nutrient sources (e.g. soil weathering, atmospheric deposition, biological fixation, and runoff from natural ecosystems) to increasingly anthropogenic sources such as agriculture, livestock production, and wastewater discharge in most African basins between 1970 and 2050. Importantly, Sub-Saharan Africa's projected population growth, potentially doubling to 1.7 billion by 2050 (Samir and Lutz 2014), coupled with urbanization, is likely to exacerbate river pollution, especially without efficient wastewater treatment. Jones *et al* (2023) identified hotspots of water quality deterioration coinciding with areas of rapid urban expansion, particularly around major urban centers and along river basins such as the Nile and Orange. Notably, while some basins like Congo, dominated by natural nutrient export (Beusen and Bouwman 2022), largely maintain concentrations below established thresholds, our models project increases by 2050 due to growing anthropogenic influences under the SSP5-RCP8.5 scenario.

Furthermore, future climate simulations of precipitation indicate increased annual sediment and nutrient (TN and TP) loads (Ockenden *et al* 2017, Qiu *et al* 2021), likely driven by enhanced runoff and flushing processes that are explicitly represented in both models. However, with the use of annual concentrations, we may miss these flushing events. River flow projections (figure S11) indicate increased precipitation in most African rivers, with significant increments in basins such as Jubba and the Sahel region. These changes are likely to result in increased surface runoff, potentially leading to higher nutrient loading that could pose a threat to fish populations. The vulnerability of freshwater fish species to climate change was highlighted by Barbarossa *et al* (2021), who found that 36% of global freshwater fish species have over half of their present-day geographical range exposed to climate extremes beyond current levels.

The proportion of African freshwater fish species with more than half of their geographical range threatened by nutrient pollution is projected to increase from 15% to 42% under IMAGE-GNM, while

SWAT+ projects over 70% of species to remain highly threatened in both 2010 and 2050 (figure 4) generally indicating an increasing trend in the vulnerability of freshwater fish species to nutrient pollution. Our results confirm previous findings on the alarming threat of extinction on freshwater fish by nutrient pollution (Miranda *et al* 2022, Zhou *et al* 2024). Elevated nutrient levels promote algal blooms which create harmful conditions for freshwater fish (Camargo and Alonso 2006, Yan *et al* 2021). These effects are corroborated by Heiskary and Markus (2003), who reported significant negative correlations between fish and macroinvertebrate assemblage characteristics and concentrations of TP and TN in nonwadeable rivers in Minnesota. Similarly, Wang *et al* (2007) observed strong relationships between nutrient concentrations and most fish assemblage measures in wadeable streams in Wisconsin. Most freshwater fish are sensitive to eutrophication, which can lead to hypoxic conditions and changes in plankton communities that affect their food webs (Bertora *et al* 2021, Poiesz *et al* 2024). Even though Anguilliformes (eels) had the least threatened areas due to nutrient pollution, they remain vulnerable to other environmental stressors such as habitat fragmentation and biological invasions (Bertora *et al* 2021), despite some tolerance to eutrophic conditions (Drouineau *et al* 2018). Our findings indicate a general increasing trend in the PAF of freshwater fish species across different ecoregions and river basins from 2010 to 2050. This trend is particularly pronounced in tropical watersheds, aligning with previous studies suggesting large climate change-induced freshwater habitat degradation in the tropics (Barbarossa *et al* 2021). Zhou *et al* (2024) showed that characterization factors for phosphorus and nitrogen impacts on freshwater fish are higher in densely populated regions encompassing either large lakes or the headwaters of large rivers. This is consistent with our observation of PAF values above 0.6 in the African Great Lakes region (figure 6).

The insights from this study can guide regional or national strategies to: (1) prioritize good practices for nutrient water quality management in rivers, including nature-based solutions such as establishing riparian buffer zones, bed and bank stabilization, as well as developing and enforcing stricter regulations on polluting industries and promoting public education and outreach, (2) implement sustainable agricultural practices to avoid further losses to the streams and rivers and (3) invest in efficient wastewater treatment methodologies and infrastructure for industry and population waste to avoid high loads of point sources to ground and surface water. Given the vulnerability of regions with rapid population growth and inadequate sanitation infrastructure to the adverse impacts of algal blooms (Suresh *et al* 2023), effective solutions will require integrated policies that align environmental management with socio-economic strategies. A unified approach in managing transboundary water resources is also important to develop consistent standards for nutrient pollution control, improve data collection and data sharing, and strengthen enforcement efforts to preserve water quality across regions (UNEP 2010). Consequently, this would contribute to achieving targets of SDG 6 (clean water and sanitation) and SDG 14 (life below water) under the condition that also SDG 2 (no hunger) is achieved.

4.3. Limitations of the study

Because the IMAGE–GNM model operates on a coarse 0.5° (∼55 km) grid, our analysis is limited to large catchments ($\geqslant$3025 km^2). This spatial limitation means that while our study provides a broad assessment of nutrient pollution, finer-scale processes such as storm-driven runoff, localized point-source discharges and riparian nutrient retention in smaller catchments may not be fully captured. Moreover, we used annual mean nutrient concentrations to assess risk largely because of the temporal resolution of most global/regional models (Strokal *et al* 2025). IMAGE–GNM operates at a yearly timestep, while SWAT+ outputs were aggregated to annual averages to enable model comparison. While this approach is suitable for assessing long-term exposure, it does not capture seasonal or short-term peak events, which can strongly influence nutrient export (Blaen *et al* 2017). Consequently, our threshold-based assessments should therefore be interpreted as first-order, broad-scale indicators of nutrient pollution risk that identify general hotspot regions rather than precise local exceedances. Future studies using higher temporal resolution models could better capture seasonal variability.

Another limitation is that our study uses generalized threshold concentrations for nutrient pollution derived from global fish metrics, which may not accurately reflect local conditions in African freshwater ecosystems. Thresholds can vary by water body type and environmental conditions (Poikane *et al* 2019), and specific toxicity thresholds for many African freshwater ecosystems remain largely unknown (de Villiers and Thiart 2007). In addition, fish species adapted to low-nutrient environments may have lower or higher thresholds for nitrogen and phosphorus toxicity. While our study uses nutrient thresholds (0.7 mg N l^{-1} for TN and 0.07 mg P l^{-1} for TP) tailored to fish metrics, these are higher than thresholds reported for other ecological indicators, such as benthic chlorophyll (Dodds *et al* 1998) and periphyton (McDowell *et al* 2020). To evaluate the robustness of our findings to threshold selection, we conducted a sensitivity analysis by varying the default thresholds by ±20%. The results showed that the

overall spatial patterns of nutrient pollution risk remained largely consistent across all threshold scenarios for both TN and TP (figures S7–S9). While the absolute extent of critical and non-critical areas changed with threshold choice, SWAT+ consistently classified a substantially larger proportion critical areas than IMAGE-GNM across all years and scenarios. Despite differences in magnitude, both models projected increasing eutrophication risk under future climate and socioeconomic conditions, suggesting that the main conclusions of our study are relatively robust to moderate threshold uncertainty. Nevertheless, future research should prioritize establishing region-specific toxicity thresholds based on biological responses of African freshwater fish species to enhance the accuracy of nutrient risk assessments and inform targeted management strategies.

A core assumption in this study is that increases in TN and TP concentrations correlate with declines in fish species richness. This assumption could be refined by increasing species observations, as current limitations may lead to overestimating species sensitivity due to underestimated richness in certain ecoregions. Access to more occurrence records could reduce uncertainty in data-scarce areas (Zhou *et al* 2023). Potential sampling bias in our point occurrence dataset presents another source of uncertainty. Many GBIF occurrence records are affected by spatial and taxonomic biases associated with uneven sampling effort, accessibility, and digitization limitations, which may not fully represent actual biodiversity patterns (Ball *et al* 2025, Bowler *et al* 2025). As a result, data gaps in fish occurrence records, particularly in regions already vulnerable to nutrient pollution, could influence the accuracy of the projected biodiversity risk patterns.

Fish species loss is influenced by a complex interplay of factors beyond nutrient pollution, including freshwater salinization (Ahmed *et al* 2022), increasing water temperatures, changing outflow (Reid *et al* 2019), and micro/nano plastics pollution (Jaiswal *et al* 2022). These factors often interact, potentially exacerbating impacts. Future research should account for additional anthropogenic pressures such as habitat fragmentation, altered river connectivity, water abstraction and overfishing (Sanon *et al* 2020). Incorporating these stressors into ecological risk assessments would provide a more comprehensive understanding of the cumulative threats to freshwater fish biodiversity and help disentangle the effects of nutrient pollution from other interacting pressures (Barbarossa *et al* 2020). The contribution of climate change to future nutrient pollution varies across Africa. For example, projected increases in river discharge in the Jubba Basin and parts of the Sahel (figure S11) may enhance nutrient transport, while hotspots in the Volta, Turkana, and Niger basins likely reflect combined hydrological and anthropogenic pressures (Chawanda *et al* 2024). Given the timescale and scope of climate change, it is unlikely that mitigation efforts will prevent all biodiversity losses, particularly in sensitive freshwater ecosystems (Richman *et al* 2015), therefore building ecological resilience through the maintenance of habitat connectivity is essential to support species distribution shifts in response to a changing climate. For many freshwater fish, maintaining natural connectivity between habitats will be vital for adapting to environmental changes and reducing species vulnerability to nutrient pollution and eutrophication (Walker *et al* 2002).

5. Conclusion

This study highlights the urgency of addressing TN and TP nutrient pollution in Africa's rivers. Using IMAGE-GNM and SWAT+ models, we have identified pollution hotspots and assessed the associated threats to freshwater fish species, providing a framework for understanding the extent and potential impact of river nutrient pollution across the continent. Our analysis shows that by 2050, under the SSP5-RCP8.5 scenario, IMAGE-GNM projects that 42% of African freshwater fish species would have at least half of their geographical range threatened by nutrient pollution, while SWAT+ projected similarly high threats for over 70% of species. Major rivers such as Niger, Volta, and Turkana will see a significant fraction (>0.8) of fish species affected. These threats are concerning as eutrophication not only degrades habitat quality but also intensifies competition, potentially leading to substantial species loss. Addressing these challenges will require a coordinated policy response including stricter regulation of nutrient emissions, sustainable agricultural practices, improved wastewater treatment, and transboundary water management. Ultimately, achieving ambient water quality and protecting fish biodiversity will demand increased public awareness, investments in pollution prevention, and a stronger commitment to cross-sectoral collaboration. Only through such comprehensive efforts can Africa's freshwater resources be preserved for future generations, safeguarding both the ecological integrity of aquatic habitats and the human communities that depend on them.

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Data availability statement

All nutrient simulations used in this study are available under the ISIMIP Water quality sector protocol (www.isimip.org/). Shapefile for freshwater ecoregions available for download at www.feow.org/download.

All data that support the findings of this study are included within the article (and any supplementary information files).

Supplementary data 1 available at: https://doi.org/10.1088/3033-4942/ae892f/data1

Supplementary data 2 available at: https://doi.org/10.1088/3033-4942/ae892f/data2

Conflict of interest

The authors declare that they have no conflict of interest.

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References

Ahmed S F, Kumar P S, Kabir M, Zuhara F T, Mehjabin A, Tasannum N, Hoang A T, Kabir Z and Mofijur M 2022 Threats, challenges and sustainable conservation strategies for freshwater biodiversity *Environ. Res.* **214** 113808

Arnold J G, Bieger K, White M J, Srinivasan R, Dunbar J A and Allen P M 2018 Use of decision tables to simulate management in SWAT+ *Water* **10** 713

Arnold J G, Srinivasan R, Muttiah R S and Williams J R 1998 Large area hydrologic modeling and assessment part I: model development[1] *JAWRA J. Am. Water Resour. Assoc.* **34** 73–89

Azevedo L B, van Zelm R, Elshout P M F, Hendriks A J, Leuven R S E W, Struijs J, de Zwart D and Huijbregts M A J 2013 Species richness–phosphorus relationships for lakes and streams worldwide *Glob. Ecol. Biogeogr.* **22** 1304–14

Ball L *et al* 2025 What 'unexplored' means: mapping regions with digitized natural history records to look for 'biodiversity blindspots *Peer J.* **13** e18511

Barbarossa V, Bosmans J, Wanders N, King H, Bierkens M F P, Huijbregts M A J and Schipper A M 2021 Threats of global warming to the world's freshwater fishes *Nat. Commun.* **12** 1701

Bertora A, Grosman F, Sanzano P and Rosso J J 2021 Longitudinal patterns in distribution of native and non-native fish species in a regulated temperate neotropical river *Acta Limnol. Bras.* 33

Beusen A H W and Bouwman A F 2022 Future projections of river nutrient export to the global coastal ocean show persisting nitrogen and phosphorus distortion *Front. Water* 4

Beusen A H W, Bouwman A F, Van Beek L P H, Mogollón J M and Middelburg J J 2015a Global riverine N and P transport to ocean increased during the twentieth century despite increased retention along the aquatic continuum *Biogeosciences* **13** 2441–51

Beusen A H W, Doelman J C, Van Beek L P H, Van Puijenbroek P J T M, Mogollón J M, Van Grinsven H J M, Stehfest E, Van Vuuren D P and Bouwman A F 2022 Exploring river nitrogen and phosphorus loading and export to global coastal waters in the shared socio-economic pathways *Glob. Environ. Change* **72** 102426

Beusen A H W, Van Beek L P H, Bouwman A F, Mogollón J M and Middelburg J J 2015b Coupling global models for hydrology and nutrient loading to simulate nitrogen and phosphorus retention in surface water—description of IMAGE–GNM and analysis of performance *Geosci. Model Dev.* **8** 4045–67

Bieger K, Arnold J G, Rathjens H, White M J, Bosch D D, Allen P M, Volk M and Srinivasan R 2017 Introduction to SWAT +, A completely restructured version of the soil and water assessment tool *JAWRA J. Am. Water Resour. Assoc.* **53** 115–30

Birk S, Chapman D, Carvalho L, Spears B M, Andersen H E, Argillier C, Auer S, Baattrup-Pedersen A, Banin L and Beklioğlu M 2020 Impacts of multiple stressors on freshwater biota across spatial scales and ecosystems *Nat. Ecol. Evol.* **4** 1060–8

Blaen P J, Khamis K, Lloyd C, Comer-warner S, Ciocca F, Thomas R M, MacKenzie A R and Krause S 2017 High-frequency monitoring of catchment nutrient exports reveals highly variable storm event responses and dynamic source zone activation *J. Geophys. Res. Biogeosci.* **122** 2265–81

Bosmans J, Wanders N, Bierkens M F P, Huijbregts M A J, Schipper A M and Barbarossa V 2022 FutureStreams, a global dataset of future streamflow and water temperature *Sci. Data* **9** 307

Bowler D E, Boyd R J, Callaghan C T, Robinson R A, Isaac N J B and Pocock M J O 2025 Treating gaps and biases in biodiversity data as a missing data problem *Biol. Rev.* **100** 50–67

Brown L C 1987 *The Enhanced Stream Water Quality Models QUAL2E and QUAL2E-UNCAS: Documentation and User Manual* (Environmental Research Laboratory. Office of Research and Development)

Camargo J A and Alonso Á 2006 Ecological and toxicological effects of inorganic nitrogen pollution in aquatic ecosystems: a global assessment *Environ. Int.* **32** 831–49

Capon S J, Stewart-Koster B and Bunn S E 2021 Future of freshwater ecosystems in a 1.5 C warmer world *Front. Environ. Sci.* **9** 784642

Chawanda C J, Nkwasa A, Thiery W and van Griensven A 2024 Combined impacts of climate and land-use change on future water resources in Africa *Hydrol. Earth Syst. Sci.* **28** 117–38

de Villiers S and Thiart C 2007 The nutrient status of South African rivers: concentrations, trends and fluxes from the 1970s to 2005 *S. Afr. J. Sci.* **103** 343–9

Desbureaux S, Mortier F, Zaveri E, van Vliet M T H, Russ J, Rodella A S and Damania R 2022 Mapping global hotspots and trends of water quality (1992–2010): a data driven approach *Environ. Res. Lett.* **17** 114048

Devlin M and Brodie J 2023 Nutrients and eutrophication *Marine Pollution–Monitoring, Management and Mitigation* (Springer) pp 75–100

Dodds W K, Jones J R and Welch E B 1998 Suggested classification of stream trophic state: distributions of temperate stream types by chlorophyll, total nitrogen, and phosphorus *Water Res.* **32** 1455–62

Drouineau H, Durif C, Castonguay M, Mateo M, Rochard E, Verreault G, Yokouchi K and Lambert P 2018 Freshwater eels: a symbol of the effects of global change *Fish Fish.* **19** 903–30

Dudgeon D and Strayer D L 2024 Bending the curve of global freshwater biodiversity loss: what are the prospects? *Biol. Rev.* **100** 205–26

EPA 2015 The facts about nutrient pollution

FEOW 2025 Freshwater ecoregions of the world (FEOW) (available at: www.Feow.Org/Background)

Gaber H S, El-kasheif M A, Ibrahim S A and Authman M 2013 Effect of water pollution in El-Rahawy drainage canal on hematology and organs of freshwater fish *World Appl. Sci. J.* **21** 329–41

Heiskary S and Markus H 2003 Establishing relations among instream nutrient concentrations, phytoplankton and periphyton abundance and composition, fish and macroinvertebrate indices, and biological oxygen demand in Minnesota USA rivers. *Final Report to USEPA Region V. Minnesota Pollution Control Agency, St. Paul, Minnesota*

Jaiswal K K, Dutta S, Banerjee I, Pohrmen C B, Singh R K, Das H T, Dubey S and Kumar V 2022 Impact of aquatic microplastics and nanoplastics pollution on ecological systems and sustainable remediation strategies of biodegradation and photodegradation *Sci. Total Environ.* **806** 151358

Jones E R, Bierkens M F P, van Puijenbroek P J T M, van Beek L P H, Wanders N, Sutanudjaja E H and van Vliet M T H 2023 Sub-Saharan Africa will increasingly become the dominant hotspot of surface water pollution *Nat. Water* **1** 602–13

Jones E R, Bierkens M F P, Wanders N, Sutanudjaja E H, van Beek L P H and van Vliet M T H 2022 Current wastewater treatment targets are insufficient to protect surface water quality *Commun. Earth Environ.* **3** 221

Kriegler E *et al* 2017 Fossil-fueled development (SSP5): an energy and resource intensive scenario for the 21st century *Glob. Environ. Change* **42** 297–315

Krishnadas H and Matysik K 2024 Water accessibility *Urban Water Ecosystems in Africa and Asia* (Routledge) pp 220–8

Liu Y, Jia Z, Ma X, Wang Y, Guan R, Guan Z, Gu Y and Zhao W 2022 Analysis of drought characteristics projections for the tibetan plateau based on the GFDL-ESM2M climate model *Remote Sens.* **14** 5084

Loarie S R, Duffy P B, Hamilton H, Asner G P, Field C B and Ackerly D D 2009 The velocity of climate change *Nature* **462** 1052–5

McDowell R W, Noble A, Pletnyakov P, Haggard B E and Mosley L M 2020 Global mapping of freshwater nutrient enrichment and periphyton growth potential *Sci. Rep.* **10** 3568

McManamay R A, Griffiths N A, DeRolph C R and Pracheil B M 2018 A synopsis of global mapping of freshwater habitats and biodiversity: implications for conservation *Pure Appl. Biogeogr.* 57–85

Miltner R J and Rankin A E T 1998 Primary nutrients and the biotic integrity of rivers and streams *Freshw. Biol.* **40** 145–58

Miranda R *et al* 2022 Monitoring extinction risk and threats of the world's fishes based on the Sampled Red List Index *Rev. Fish Biol. Fish.* **32** 975–91

Mustafa S A, Al-rudainy A J and Salman N M 2024 Effect of environmental pollutants on fish health: an overview *Egypt. J. Aquat. Res.* **50** 225–33

Neitsch S L, Arnold J G, Kiniry J R and Williams J R 2011 *Soil and Water Assessment Tool Theoretical Documentation Version 2009* (Texas Water Resources Institute)

Neitsch S L, Arnold J G, Kiniry J R, Williams J R and King K W 2005 SWAT theoretical documentation version 2005 *Soil and Water Research Laboratory* (*ARS*) pp 1–494

Nkwasa A, James Chawanda C, Theresa Nakkazi M, Tang T, Eisenreich S J, Warner S and van Griensven A 2024 One third of African rivers fail to meet the'good ambient water quality' nutrient targets *Ecol. Indic.* **166** 112544

Nyenje P M, Foppen J W, Uhlenbrook S, Kulabako R and Muwanga A 2010 Eutrophication and nutrient release in urban areas of sub-Saharan Africa—a review *Sci. Total Environ.* **408** 447–55

O'Neill B C *et al* 2017 The roads ahead: narratives for shared socioeconomic pathways describing world futures in the 21st century *Glob. Environ. Change* **42** 169–80

Ockenden M C *et al* 2017 Major agricultural changes required to mitigate phosphorus losses under climate change *Nat. Commun.* **8** 161

Pérez C J, Vega-rodríguez M A, Reder K and Flörke M 2017 A multi-objective artificial bee colony-based optimization approach to design water quality monitoring networks in river basins *J. Clean. Prod.* **166** 579–89

Poiesz S S H, Witte J I J and van der Veer H W 2024 Stomach content analysis indicates multi decadal trophic stability in a temperate coastal fish food web, western Dutch Wadden sea *Estuarine, Coastal and Shelf Sci.* **308** 108912

Poikane S, Kelly M G, Salas Herrero F, Pitt J-A, Jarvie H P, Claussen U, Leujak W, Lyche Solheim A, Teixeira H and Phillips G 2019 Nutrient criteria for surface waters under the European water framework directive: current state-of-the-art, challenges and future outlook *Sci. Total Environ.* **695** 133888
Poikane S, Várbíró G, Kelly M G, Birk S and Phillips G 2021 Estimating river nutrient concentrations consistent with good ecological condition: more stringent nutrient thresholds needed *Ecol. Indic.* **121** 107017
Qiu J, Shen Z, Leng G and Wei G 2021 Synergistic effect of drought and rainfall events of different patterns on watershed systems *Sci. Rep.* **11** 18957
Radinger J, Essl F, Hölker F, Horký P, Slavík O and Wolter C 2017 The future distribution of river fish: the complex interplay of climate and land use changes, species dispersal and movement barriers *Glob. Change Biol.* **23** 4970–86
Reid A J *et al* 2019 Emerging threats and persistent conservation challenges for freshwater biodiversity *Biol. Rev.* **94** 849–73
Riahi K, Rao S, Krey V, Cho C, Chirkov V, Fischer G, Kindermann G, Nakicenovic N and Rafaj P 2011 RCP 8.5—a scenario of comparatively high greenhouse gas emissions *Clim. Change* **109** 33–57
Robertson D M 2006 *Nutrient Concentrations and Their Relations to the Biotic Integrity of Wadeable Streams in Wisconsin* (US Department of the Interior, US Geological Survey)
Sabater S, Freixa A, Jiménez L, López-Doval J, Pace G, Pascoal C, Perujo N, Craven D and González-trujillo J D 2023 Extreme weather events threaten biodiversity and functions of river ecosystems: evidence from a meta-analysis *Biol. Rev.* **98** 450–61
Samir K C and Lutz W 2014 The human core of the shared socioeconomic pathways: population scenarios by age, sex and level of education for all countries to 2100 *Glob. Environ. Change* **42** 181–92
Sanon V P *et al* 2020 Multiple-line identification of socio-ecological stressors affecting aquatic ecosystems in semi-arid countries: implications for sustainable management of fisheries in Sub-Saharan Africa *Water* **12** 1518
Sayer C A *et al* 2025 One-quarter of freshwater fauna threatened with extinction *Nature* **638** 138–45
Siabi E K, Awafo E A, Kabo-bah A T, Derkyi N S A, Akpoti K, Mortey E M and Yazdanie M 2023 Assessment of Shared Socioeconomic Pathway (SSP) climate scenarios and its impacts on the Greater Accra region *Urban Clim.* **49** 101432
Sierra J P, Arias P A and Vieira S C 2015 Precipitation over Northern South America and its seasonal variability as simulated by the CMIP5 models *Adv. Meteorol.* **2015** 1–22
Snoeks J and Theeten F 2025 African fish collection of the Africamuseum. Version 1.22. royal museum for central Africa, Belgium ((available at: GBIF.Org)
Soro M-P, N'goran K M, Ouattara A A, Yao K M, Kouassi N L B and Diaco T 2023 Nitrogen and phosphorus spatio-temporal distribution and fluxes intensifying eutrophication in three tropical rivers of Côte d'Ivoire (West Africa) *Mar. Pollut. Bull.* **186** 114391
Strokal M *et al* 2024 The water quality protocol for model intercomparisons under climate change impacts *Eur. Geosciences Union General Assembly* **2024** 2180
Strokal M *et al* 2025 Advancing water quality model intercomparisons under global change: perspectives from the new ISIMIP water quality sector *Environ. Res. Water* **1** 035002
Suresh K, Tang T, van Vliet M T H, Bierkens M F P, Strokal M, Sorger-Domenigg F and Wada Y 2023 Recent advancement in water quality indicators for eutrophication in global freshwater lakes *Environ. Res. Lett.* **18** 063004
Sutanudjaja E H, Van Beek R, Wanders N, Wada Y, Bosmans J H C, Drost N, Van Der Ent R J, De Graaf I E M, Hoch J M and De Jong K 2018 PCR-GLOBWB 2: a 5 arcmin global hydrological and water resources model *Geosci. Model Dev.* **11** 2429–53
Sutton M A, Bleeker A, Howard C M, Erisman J W, Abrol Y P, Bekunda M, Datta A, Davidson E, De Vries W and Oenema O 2013 *Our Nutrient World. The Challenge to Produce More Food & Energy with Less Pollution* (Centre for Ecology & Hydrology)
Temesgen M, Getahun A and Lemma B 2019 Livelihood functions of capture fisheries in Sub-Saharan Africa: food security, nutritional, and economic implications *Rev. Fish. Sci. Aquac.* **27** 215–25
Thomas M F and Whittington G W 2023 *Environment and Land Use in Africa* (Taylor & Francis)
Tilak K S, Veeraiah K and Raju J M P 2007 Effects of ammonia, nitrite and nitrate on hemoglobin content and oxygen consumption of freshwater fish, Cyprinus carpio (Linnaeus) *J. Environ. Biol.* **28** 45–47
UNEP 2010 Chapter 2: transboundary water resources *Africa Water Atlas. Division of Early Warning and Assessment (DEWA)* (United Nations Environment Programme (UNEP))
Valdes P J *et al* 2017 The BRIDGE HadCM3 family of climate models: hadCM3@Bristol v1.0 *Geosci. Model Dev.* **10** 3715–43
van Vliet M T H *et al* 2023 Global river water quality under climate change and hydroclimatic extremes *Nat. Rev. Earth Environ.* **4** 687–702
van Vliet M T *et al* 2019 Model inter-comparison design for large-scale water quality models *Curr. Opin. Environ. Sustain.* **36** 59–67
van Vuuren D P *et al* 2011 The representative concentration pathways: an overview *Clim. Change* **109** 5–31
Villacorte L O, Tabatabai S A A, Dhakal N, Amy G, Schippers J C and Kennedy M D 2015 Algal blooms: an emerging threat to seawater reverse osmosis desalination *Desalin. Water Treat.* **55** 2601–11
Villéger S, Brosse S, Mouchet M, Mouillot D and Vanni M J 2017 Functional ecology of fish: current approaches and future challenges *Aquat. Sci.* **79** 783–801
Wagenaar G and Barnhoorn I 2018 Health and chemical burdens of fish species from polluted and hyper-eutrophic freshwater ecosystems in South Africa *Afr. J. Aquat. Sci.* **43** 271–80
Wang L, Robertson D M and Garrison P J 2007 Linkages between nutrients and assemblages of macroinvertebrates and fish in wadeable streams: implication to nutrient criteria development *Environ. Manag.* **39** 194–212
Wang M, Bodirsky B L, Rijneveld R, Beier F, Bak M P, Batool M, Droppers B, Popp A, van Vliet M T H and Strokal M 2024 A triple increase in global river basins with water scarcity due to future pollution *Nat. Commun.* **15** 880
Westlund L, Holvoet K and Kébé M 2008 *Achieving poverty reduction through responsible fisheries: lessons from West and Central Africa*
Whitehead P G, Wilby R L, Battarbee R W, Kernan M and Wade A J 2009 A review of the potential impacts of climate change on surface water quality *Hydrol. Sci. J.* **54** 101–23
Yan Q, Cheng T, Song J, Zhou J, Hung C-C and Cai Z 2021 Internal nutrient loading is a potential source of eutrophication in Shenzhen Bay, China *Ecol. Indic.* **127** 107736
Yasin J A, Kroeze C and Mayorga E 2010 Nutrients export by rivers to the coastal waters of Africa: past and future trends *Glob. Biogeochem. Cycles* **24**
Zhou J, Mogollón J M, van Bodegom P M, Barbarossa V, Beusen A H W and Scherer L 2023 Effects of nitrogen emissions on fish species richness across the world's freshwater ecoregions *Environ. Sci. Technol.* **57** 8347–54
Zhou J, Mogollón J M, van Bodegom P M, Beusen A H W and Scherer L 2024 Global regionalized characterization factors for phosphorus and nitrogen impacts on freshwater fish biodiversity *Sci. Total Environ.* **912** 169108